

Whole-Brain Dendrite Mesh Proximity and Synaptic Connectivity

FlyWire v783 proximity analysis

June 21, 2026

Abstract

This report quantifies how minimum dendrite mesh distance relates to synaptic connectivity in the FlyWire v783 connectome. The analysis includes 139,255 neurons and 77,523,908 neuron pairs whose sampled dendrite meshes are within 2 microns. Of these near pairs, 12,813,758 have at least one synaptic connection in either direction, giving an overall connected fraction of 16.53%. The dominant quantitative rule is distance decay: the 0 to 500 nm window has fraction connected 31.3%, while the 1.5 to 2.0 micron window has fraction connected 1.15%.

1 Methods

The input to the analysis is the completed default whole-connectome proximity run. The manifest records the exact run name and parameter set. Meshes were fetched from the public `precomputed://gs://flywire_v141_m783` CloudVolume source, so the analysis does not require a private FlyWire or CAVE authorization token. The report uses the final enriched pair table generated by the pipeline, not a sampled subset, unless a smoke-test limit is explicitly supplied on the command line.

The dendrite mesh proxy is constructed from mesh vertices near postsynaptic sites. Vertices are sampled at 250 nm spacing after restricting to a 500 nm radius around postsynaptic coordinates. A neuron pair is counted as spatially proximal when the closest sampled dendrite mesh points are within 2 microns. The whole-brain run uses exact pair aggregation across bucketed reduce tasks, with 64 reduce buckets in the final reduce stage.

Connectivity is measured as a binary indicator of whether either direction contains at least one synapse. For distance curves, pairs are binned into 100 nm distance intervals from 0 to 2 microns. Each bin reports a connected fraction and a Wilson 95 percent confidence interval. For metadata partitions, the report compares observed connected counts with expected connected counts obtained by applying the global distance-bin connection rate to the partition-specific distance composition. This creates a distance-adjusted observed over expected statistic, reported as log2 observed over expected.

Two compact distance rules are fit to the binned data. The first is a weighted log-linear model for the connection fraction, which summarizes multiplicative decay of probability with distance. The second is a weighted logistic-binomial approximation, which summarizes multiplicative decay of connection odds with distance. These models are descriptive summaries of the binned observations. They are used to quantify effect sizes, not to claim that distance alone is a full biological mechanism.

2 Results

2.1 Global Distance Rule

The global distance curve gives the main quantitative answer. Among all near pairs within 2 microns, 16.53% are connected. The nearest 100 nm bin has fraction connected 44.7% across 18,607,001 near pairs. The farthest 100 nm bin has fraction connected 0.89% across 2,351,312 near pairs. The absolute drop from first to last bin is 43.8 percentage points.

Aggregating bins into wider windows makes the rule easier to use. The 0 to 500 nm window has fraction connected 31.3%, while the 1.5 to 2.0 micron window has fraction connected 1.15%. This is an absolute difference of 30.2 percentage points and a fold change of 27.2. The first 100 nm bin at or below 10 percent connected is 500-600 nm, and the first bin at or below 5 percent connected is 800-900 nm.

Table 1: Connection probability and connection strength across four prespecified distance windows.

Distance window	Near pairs	Connected	Fraction	95% CI	Mean synapses
0.0-0.5 um	36,524,440	11,446,028	0.313	0.313 to 0.314	4.51
0.5-1.0 um	15,481,622	914,365	0.059	0.059 to 0.059	1.76
1.0-1.5 um	13,422,444	314,131	0.023	0.023 to 0.023	1.67
1.5-2.0 um	12,095,402	139,234	0.012	0.011 to 0.012	1.66

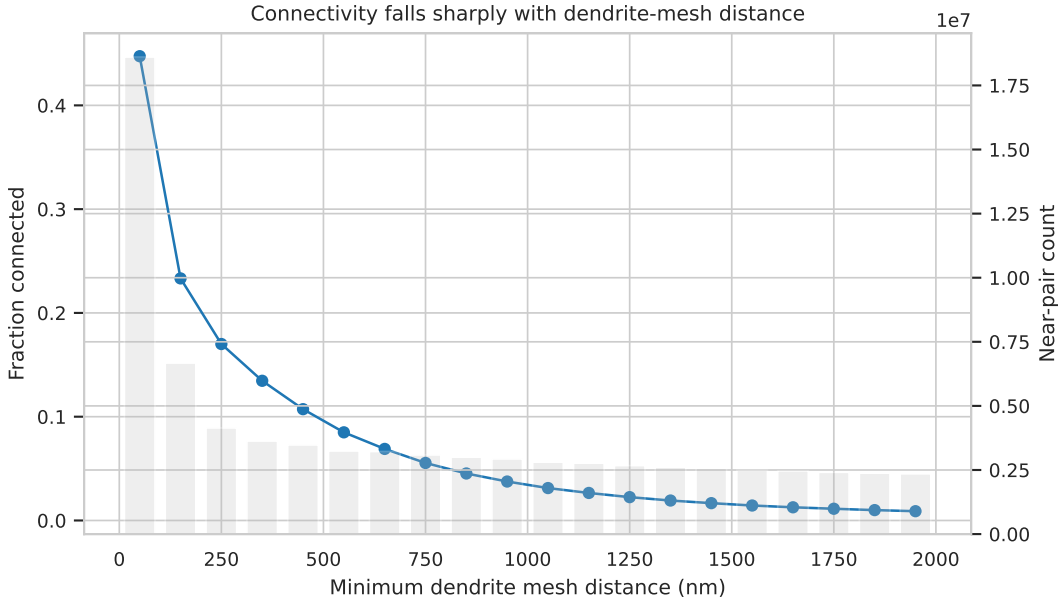


Figure 1: The blue line shows the observed fraction of neuron pairs with any synaptic connection in each 100 nm minimum dendrite mesh distance bin. The shaded band is the Wilson 95% confidence interval for each bin, and the gray bars show the number of near pairs contributing to the estimate. The main quantitative takeaway is a drop from 44.7% in the 0-100 nm bin to 0.89% in the 1900-2000 nm bin. When bins are aggregated, the 0 to 500 nm window has fraction connected 31.3%, whereas the 1.5 to 2.0 micron window has fraction connected 1.15%, a 27.2 fold change.

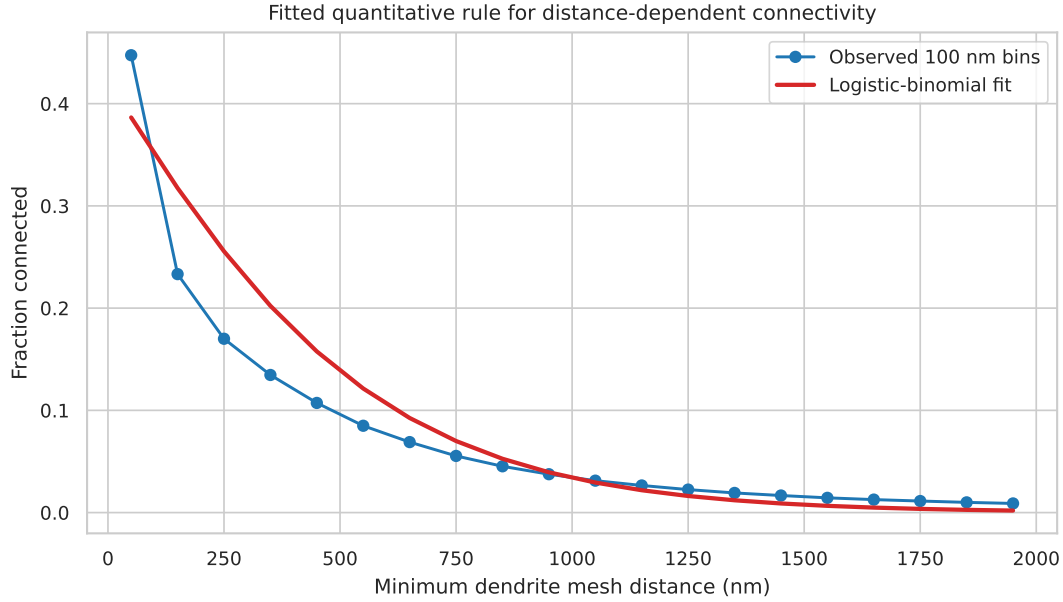


Figure 2: Points show the observed 100 nm bin fractions and the red curve shows a logistic-binomial fit to the same binned counts. This fit is used as a compact quantitative rule, not as a mechanistic model. The fitted odds ratio per additional 100 nm is 0.738, equivalent to an odds ratio of 0.048 per micron. The fitted half-rate distance is 0.37 microns, and the McFadden-style pseudo R squared is 0.219.

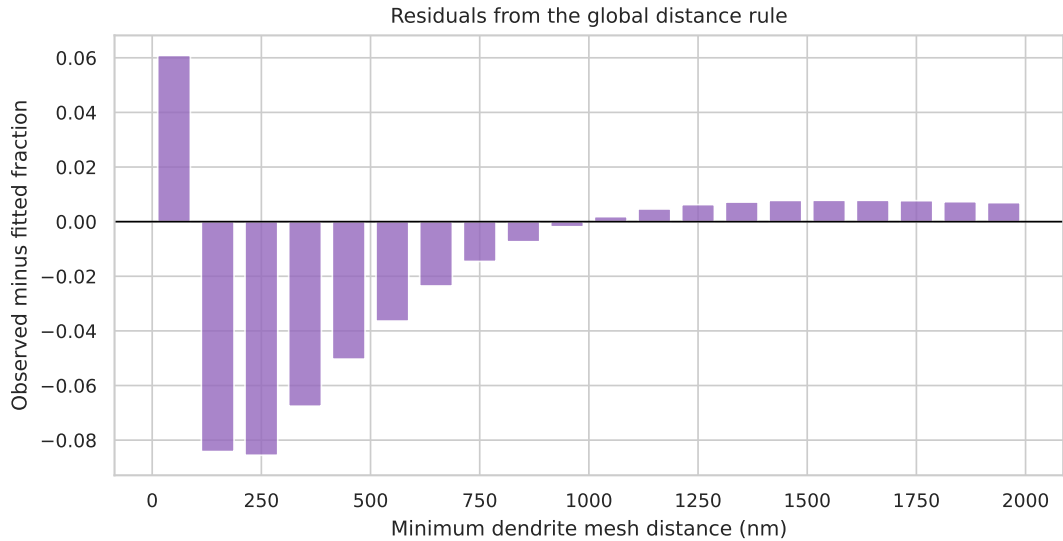


Figure 3: Bars show observed minus fitted connection fraction for each distance bin. Positive residuals indicate bins where the global distance rule underpredicts connectivity, and negative residuals indicate overprediction. The maximum absolute residual across the 20 bins is 8.55 percentage points, which provides a scale for interpreting where the one-dimensional distance rule is insufficient.

The fitted logistic-binomial rule estimates that each additional 100 nm multiplies connection odds by 0.738. Equivalently, each additional micron multiplies odds by 0.048. The weighted log-linear model gives a probability multiplier of 0.120 per micron and a probability half-distance of 0.33 microns. The logistic pseudo R squared is 0.219, and the weighted R squared on binned fractions is 0.920.

2.2 Connection Strength Among Connected Pairs

The binary connection rule asks whether a pair is connected at all. The strength analysis asks a separate question: among connected near pairs, how many synapses are present? The nearest distance bin has mean 5.38 synapses per connected pair, while the farthest bin has mean 1.67 synapses per connected pair. This comparison is conditional on being connected, so it should not be interpreted as the total expected synapse count for arbitrary near pairs.

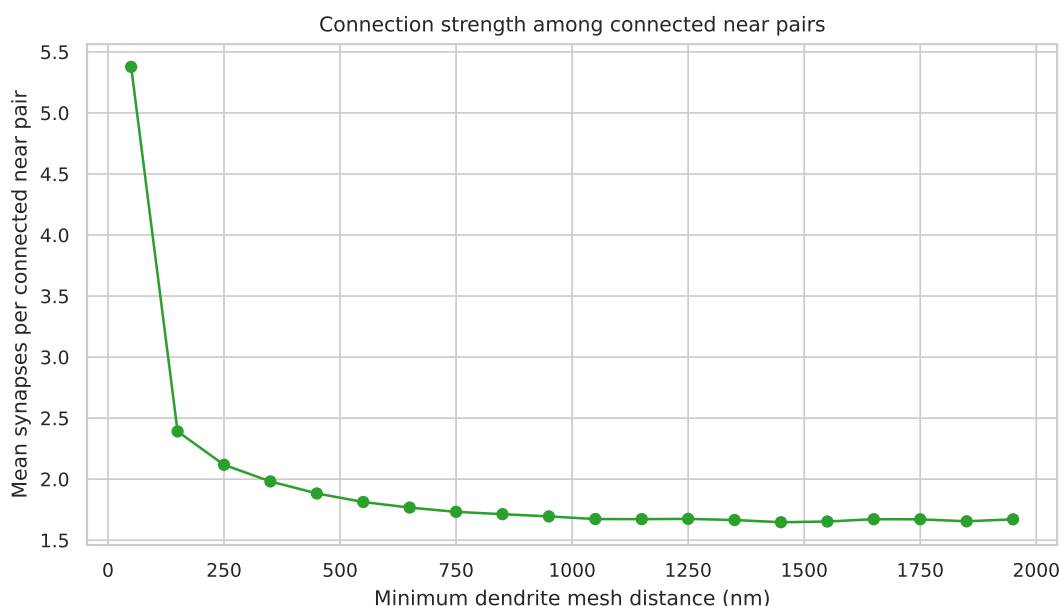


Figure 4: The curve shows the mean total synapse count among pairs that are connected, grouped by the same 100 nm distance bins. This figure separates connection probability from connection strength among successful connections. The nearest bin has mean strength 5.38 synapses per connected pair, whereas the farthest bin has mean strength 1.67 synapses per connected pair.

2.3 Broad Neuron Classes

For broad neuron classes, the selected high-support pairs cover 77,464,215 near pairs and 12,805,154 connected pairs. The median distance-adjusted effect is $\log_2$ observed over expected 0.27, with an interquartile range of 1.03. The largest positive effect is ascending | visual_projection, with $\log_2$ observed over expected 1.06 and fraction connected 20.0% across 16,685 near pairs. The largest negative effect is descending | visual_centrifugal, with $\log_2$ observed over expected -1.15 and fraction connected 7.7% across 70,958 near pairs.

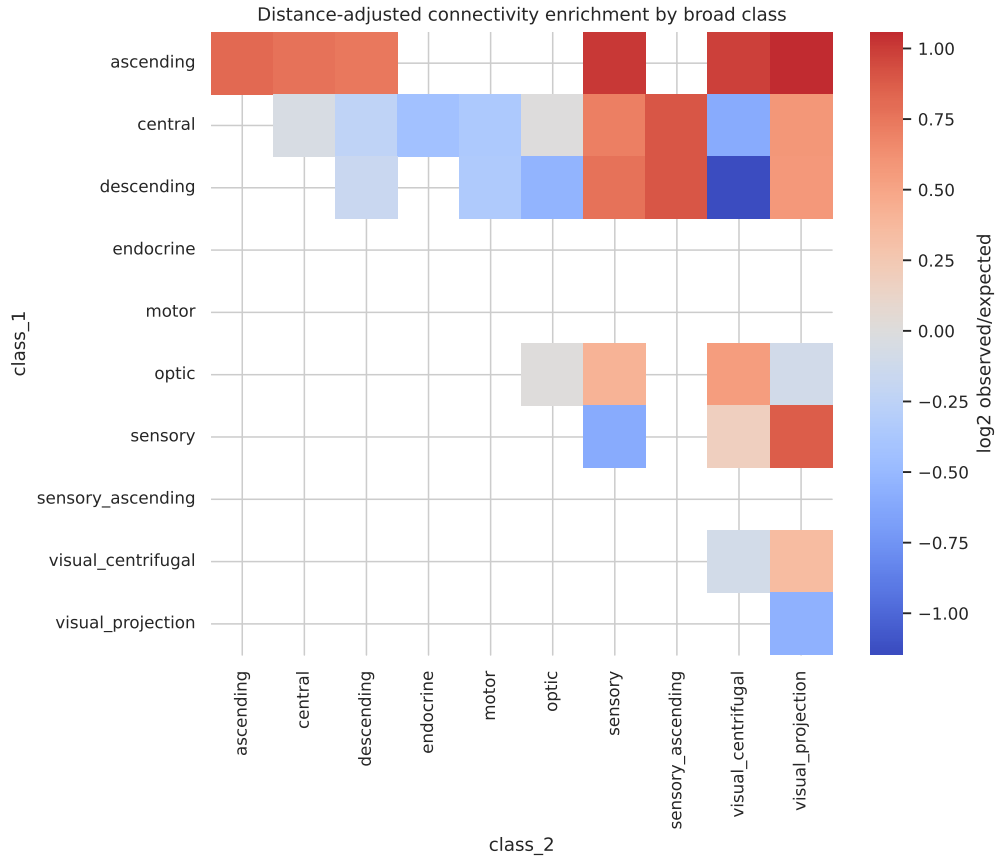


Figure 5: Cells show $\log_2$ observed over expected connectivity for broad neuron class pairs after adjusting for the global distance distribution. Red values have more connected pairs than expected from distance alone, blue values have fewer, and values near zero are close to the distance-only expectation. The strongest selected broad-class enrichment is ascending | visual_projection with $\log_2$ observed over expected 1.06.

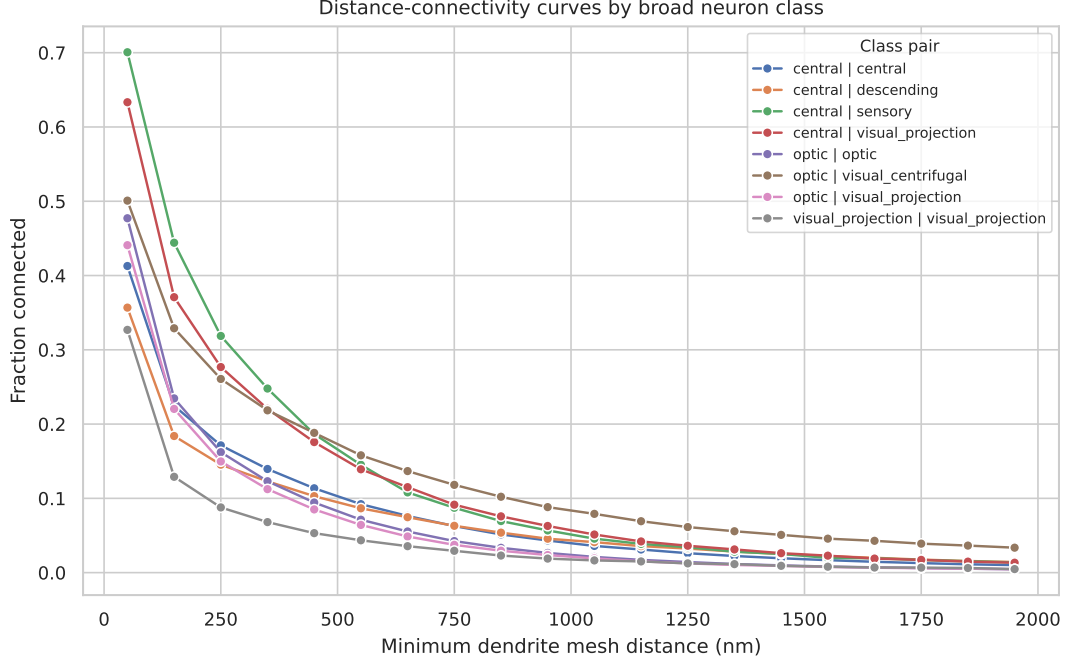


Figure 6: Each line is a distance-connectivity curve for one of the highest-support broad neuron class pairs. The comparison shows that the distance rule is shared across broad classes, while the vertical separation of curves quantifies class-specific residual differences. The selected broad-class pairs cover 77,464,215 near pairs.

2.4 Cell Classes and Exact Neuron Types

For cell classes, the selected high-support pairs cover 28,581,347 near pairs and 4,899,104 connected pairs. The median distance-adjusted effect is $\log_2$ observed over expected 0.11, with an interquartile range of 1.24. The largest positive effect is LA | visual, with $\log_2$ observed over expected 1.53 and fraction connected 32.5% across 13,972 near pairs. The largest negative effect is LO>LOP | ME>LOP.LO, with $\log_2$ observed over expected -2.64 and fraction connected 1.1% across 8,486 near pairs.

For exact neuron types, the selected high-support pairs cover 4,948,340 near pairs and 569,600 connected pairs. The median distance-adjusted effect is $\log_2$ observed over expected -0.60, with an interquartile range of 4.66. The largest positive effect is T2 | cL21, with $\log_2$ observed over expected 2.14 and fraction connected 97.5% across 2,776 near pairs. The largest negative effect is Lawf2 | Tm9, with $\log_2$ observed over expected -5.29 and fraction connected 0.3% across 6,827 near pairs.

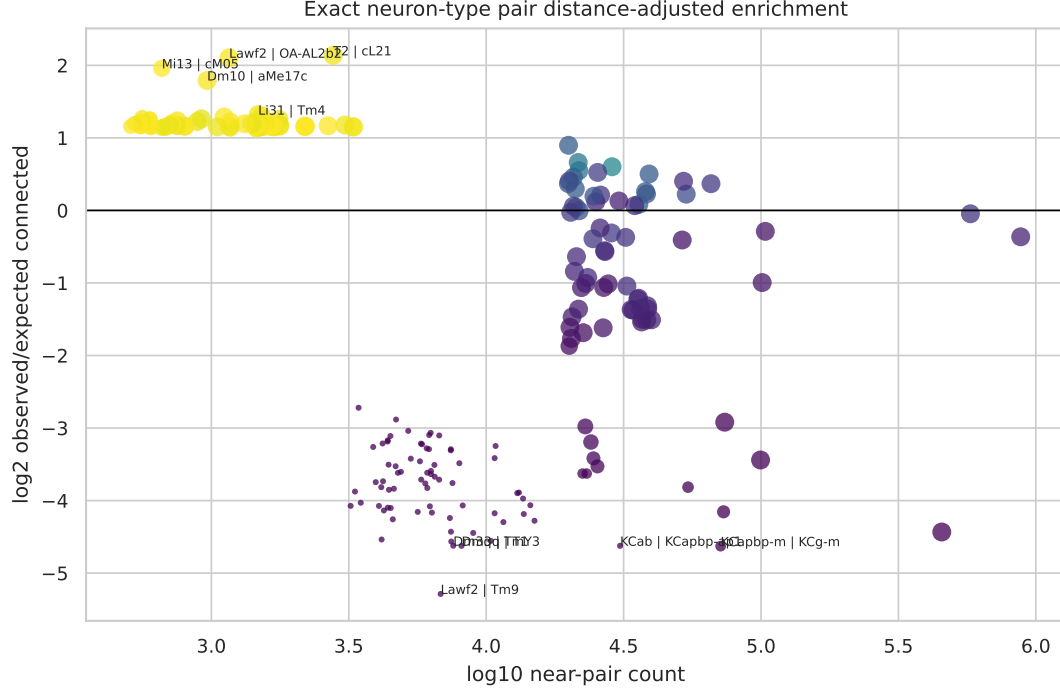


Figure 7: Each point is a selected exact neuron-type pair. The x-axis is $\log_{10}$ near-pair support, the y-axis is distance-adjusted $\log_2$ observed over expected connectivity, point color is raw fraction connected, and point area increases with connected-pair count. This representation distinguishes high-confidence effects from small-support outliers. The strongest selected exact-type enrichment is T2 | cL21, with $\log_2$ observed over expected 2.14.

2.5 Brain Regions

For dominant postsynaptic neuropils, the selected high-support pairs cover 59,918,001 near pairs and 10,241,532 connected pairs. The median distance-adjusted effect is $\log_2$ observed over expected -0.06, with an interquartile range of 1.06. The largest positive effect is LOP_L | LOP_R, with $\log_2$ observed over expected 1.56 and fraction connected 33.1% across 6,906 near pairs. The largest negative effect is AVL_P_L | MB_CA_L, with $\log_2$ observed over expected -2.43 and fraction connected 0.4% across 8,809 near pairs.

For dominant presynaptic neuropils, the selected high-support pairs cover 54,670,227 near pairs and 9,602,716 connected pairs. The median distance-adjusted effect is $\log_2$ observed over expected -0.04, with an interquartile range of 1.72. The largest positive effect is MB_CA_R | MB_ML_R, with $\log_2$ observed over expected 0.99 and fraction connected 34.3% across 19,166 near pairs. The largest negative effect is IB_L | MB_ML_L, with $\log_2$ observed over expected -4.11 and fraction connected 0.4% across 2,239 near pairs.

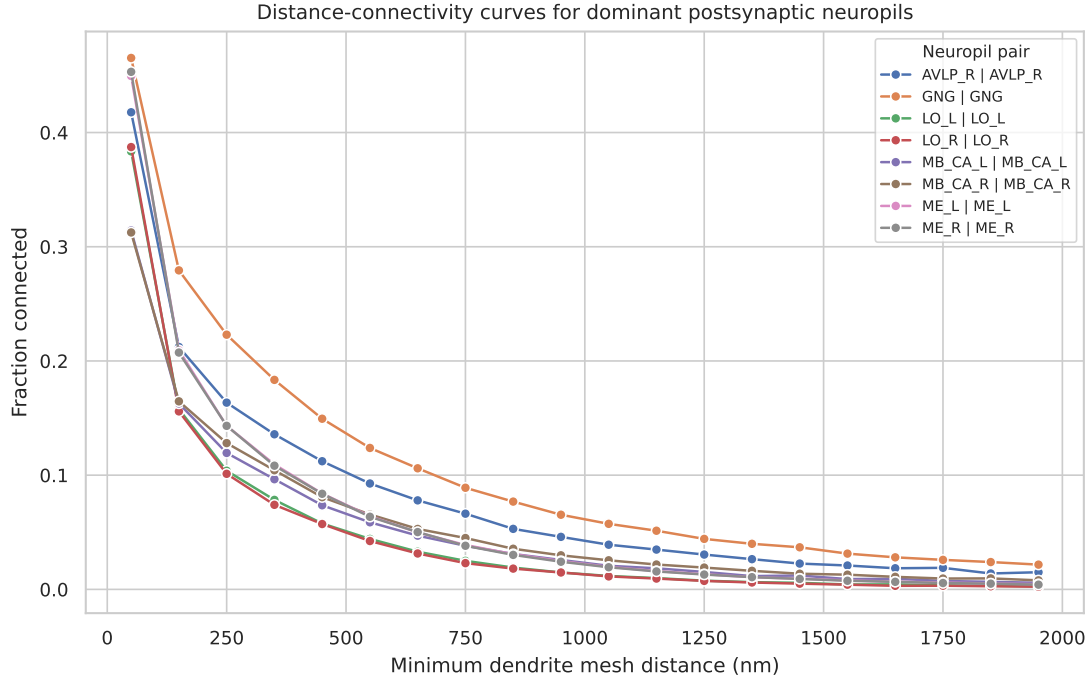


Figure 8: Lines show distance-connectivity curves for high-support same-region pairs defined by dominant postsynaptic neuropil. Each point is a 100 nm bin fraction, so differences between curves indicate region-specific connectivity beyond the global distance trend. The selected postsynaptic neuropil pairs cover 59,918,001 near pairs.

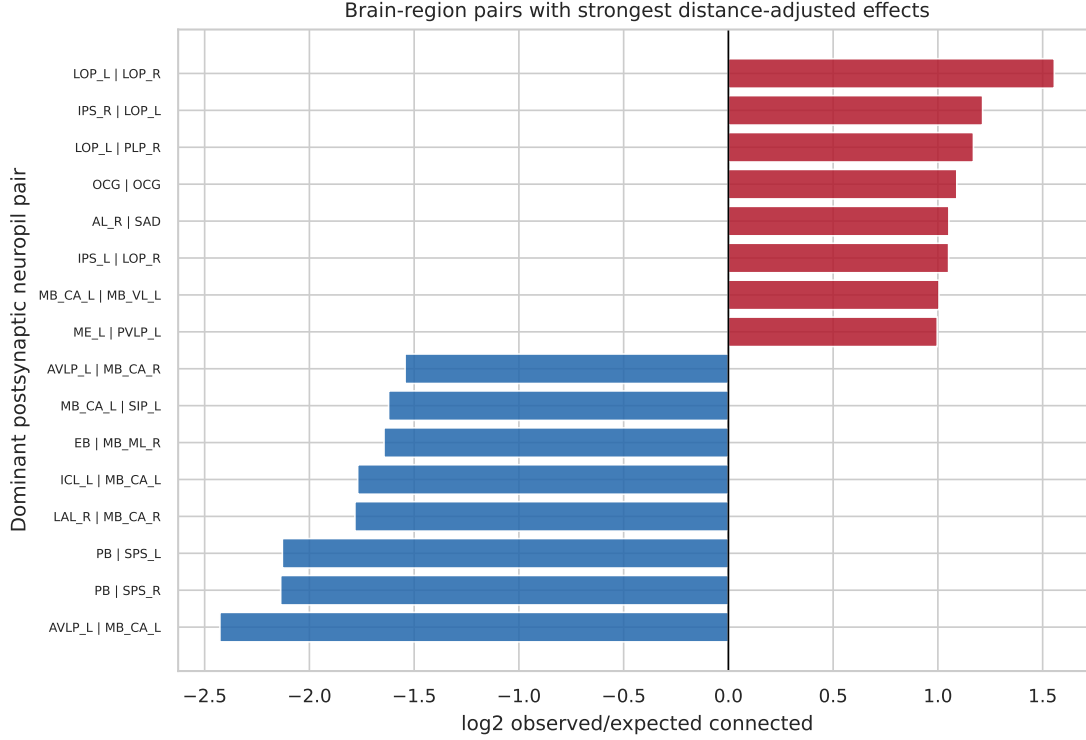


Figure 9: Bars show the largest positive and negative distance-adjusted effects for dominant postsynaptic neuropil pairs. Positive bars are pairs with more connected pairs than predicted by distance composition, and negative bars are pairs with fewer. The top enriched selected region pair is LOP_L | LOP_R, with log2 observed over expected 1.56. The strongest depletion is AVLP_L | MB_CA_L, with log2 observed over expected -2.43.

2.6 Neurotransmitter, Flow, and Side

For canonical neurotransmitter pairs, the selected high-support pairs cover 77,498,792 near pairs and 12,811,761 connected pairs. The median distance-adjusted effect is log2 observed over expected -0.00, with an interquartile range of 0.22. The largest positive effect is gaba | serotonin, with log2 observed over expected 0.25 and fraction connected 17.0% across 165,608 near pairs. The largest negative effect is octopamine | octopamine, with log2 observed over expected -0.98 and fraction connected 16.8% across 1,091 near pairs.

For flow labels, the selected high-support pairs cover 77,514,353 near pairs and 12,812,504 connected pairs. The median distance-adjusted effect is log2 observed over expected 0.01, with an interquartile range of 0.69. The largest positive effect is afferent | efferent, with log2 observed over expected 0.75 and fraction connected 20.8% across 322,683 near pairs. The largest negative effect is efferent | intrinsic, with log2 observed over expected -0.23 and fraction connected 14.6% across 2,146,760 near pairs.

For hemisphere side labels, the selected high-support pairs cover 77,514,283 near pairs and 12,812,503 connected pairs. The median distance-adjusted effect is log2 observed over expected 0.20, with an interquartile range of 0.36. The largest positive effect is na | right, with log2 observed over expected 0.42 and fraction connected 18.3% across 2,702 near pairs. The largest negative effect is left | left, with log2 observed over expected -0.05 and fraction connected 16.1% across 34,000,292

near pairs.

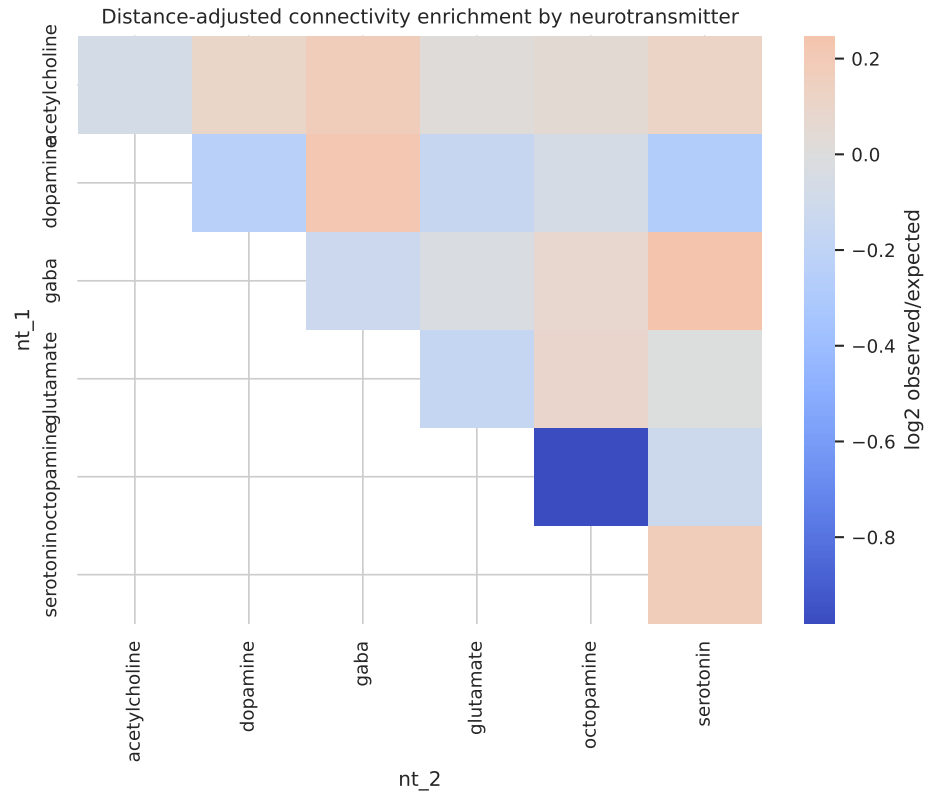


Figure 10: Cells show the same distance-adjusted observed over expected statistic after grouping each pair by canonical neurotransmitter labels. The narrower color range relative to some anatomical partitions means that broad neurotransmitter identity explains less residual variation after distance adjustment. The strongest selected neurotransmitter enrichment is gaba | serotonin, with log2 observed over expected 0.25.

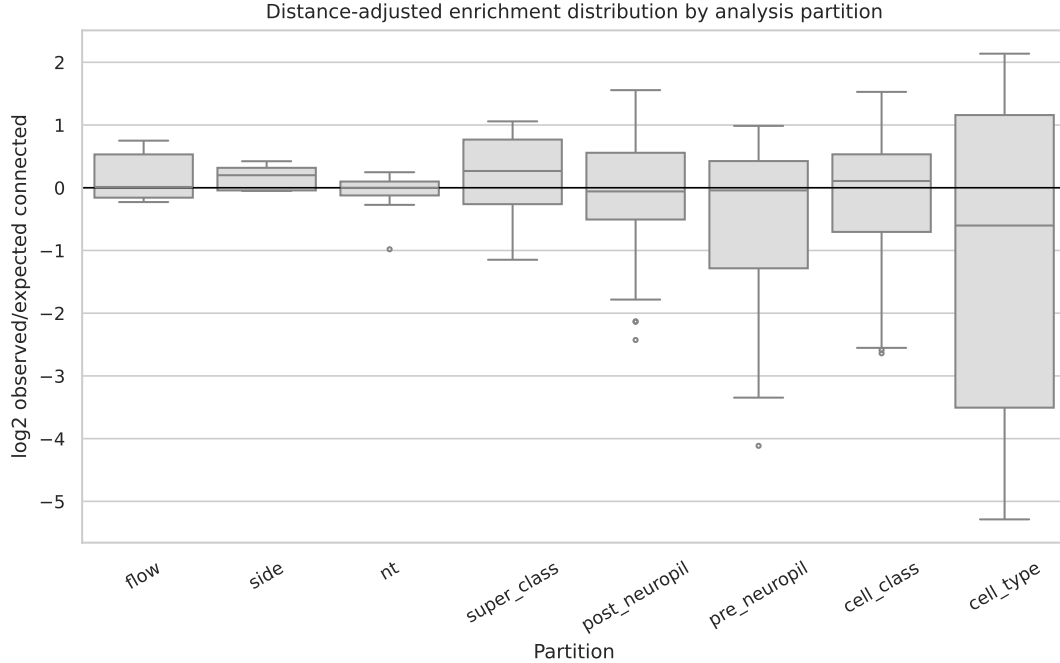


Figure 11: Boxes summarize the distribution of distance-adjusted $\log_2$ observed over expected effects among selected high-support pairs for each partition. The zero line is the distance-only expectation. Wider boxes and longer tails identify partitions where metadata labels capture more residual structure after controlling for distance. In this selected set, exact neuron type spans from -5.29 to 2.14 in $\log_2$ observed over expected.

2.7 Distance-Adjusted Effect Tables

Table 2: Broad neuron class pairs with the largest positive and negative distance-adjusted effects among selected supported pairs.

Direction	Pair	Near pairs	Connected	Fraction	Expected	$\log_2(O/E)$
enriched	ascending visual_projection	16,685	3,343	0.200	1605.1	1.06
enriched	ascending sensory	81,435	21,991	0.270	10899.0	1.01
enriched	ascending visual_centrifugal	19,359	3,953	0.204	1999.1	0.98
enriched	central sensory_ascending	46,781	7,542	0.161	4040.1	0.90
enriched	descending sensory_ascending	11,373	1,611	0.142	864.2	0.90
depleted	descending visual_centrifugal	70,958	5,460	0.077	12100.1	-1.15
depleted	sensory sensory	282,980	28,073	0.099	42671.1	-0.60
depleted	central visual_centrifugal	691,153	71,926	0.104	108960.8	-0.60
depleted	visual_projection visual_projection	2,006,829	290,967	0.145	429035.9	-0.56
depleted	descending optic	67,108	4,619	0.069	6709.4	-0.54

3 Discussion

The most defensible quantitative rule from this analysis is that dendrite mesh proximity is a powerful but incomplete predictor of synaptic connectivity. The evidence for the rule is the monotonic

Table 3: Cell class pairs with the largest positive and negative distance-adjusted effects among selected supported pairs.

Direction	Pair	Near pairs	Connected	Fraction	Expected	$\log_2(O/E)$
enriched	LA visual	13,972	4,539	0.325	1573.6	1.53
enriched	Kenyon_Cell MBIN	10,691	10,399	0.973	4658.9	1.16
enriched	LA>ME visual	79,522	24,975	0.314	11726.6	1.09
enriched	AN mechanosensory	72,955	20,489	0.281	10020.6	1.03
enriched	Kenyon_Cell MBON	121,933	62,664	0.514	31450.1	0.99
depleted	LO>LOP ME>LOP.LO	8,486	95	0.011	594.5	-2.64
depleted	ME>LO.LOP visual	40,829	233	0.006	1402.9	-2.59
depleted	ME>LOP ME>LOP.LO	6,515	67	0.010	395.5	-2.55
depleted	ME>LOP.LO visual	5,721	67	0.012	337.1	-2.32
depleted	ME>LA ME>LO.LOP	240,838	8,228	0.034	35133.1	-2.09

Table 4: Dominant postsynaptic neuropil pairs with the largest positive and negative distance-adjusted effects among selected supported pairs.

Direction	Pair	Near pairs	Connected	Fraction	Expected	$\log_2(O/E)$
enriched	LOP_L LOP_R	6,906	2,287	0.331	777.5	1.56
enriched	IPS_R LOP_L	5,316	1,643	0.309	708.2	1.21
enriched	LOP_L PLP_R	2,927	805	0.275	357.5	1.17
enriched	OCG OCG	4,033	1,334	0.331	626.2	1.09
enriched	AL_R SAD	2,909	807	0.277	388.8	1.05
depleted	AVLP_L MB_CA_L	8,809	39	0.004	212.0	-2.43
depleted	PB SPS_R	3,685	71	0.019	314.0	-2.14
depleted	PB SPS_L	3,660	66	0.018	290.4	-2.13
depleted	LAL_R MB_CA_R	4,110	32	0.008	111.3	-1.78
depleted	ICL_L MB_CA_L	5,048	38	0.008	130.8	-1.77

Table 5: Dominant presynaptic neuropil pairs with the largest positive and negative distance-adjusted effects among selected supported pairs.

Direction	Pair	Near pairs	Connected	Fraction	Expected	$\log_2(O/E)$
enriched	MB_CA_R MB_ML_R	19,166	6,568	0.343	3315.2	0.99
enriched	LA_L LA_L	25,255	6,794	0.269	3470.0	0.97
enriched	AL_R MB_CA_R	4,132	1,972	0.477	1048.8	0.91
enriched	AL_L MB_CA_L	4,397	2,001	0.455	1065.8	0.91
enriched	PB PB	3,568	1,713	0.480	913.2	0.91
depleted	IB_L MB_ML_L	2,239	8	0.004	146.7	-4.11
depleted	ATL_L MB_ML_L	2,808	18	0.006	187.6	-3.35
depleted	ICL_R MB_ML_R	8,450	30	0.004	287.4	-3.24
depleted	LA_L LOP_L	3,806	52	0.014	381.7	-2.86
depleted	LA_R LOP_R	8,318	135	0.016	932.3	-2.78

Table 6: Neurotransmitter pairs with the largest positive and negative distance-adjusted effects among selected supported pairs.

Direction	Pair	Near pairs	Connected	Fraction	Expected	$\log_2(O/E)$
enriched	gaba serotonin	165,608	28,181	0.170	23739.6	0.25
enriched	dopamine gaba	381,423	71,731	0.188	61267.7	0.23
enriched	serotonin serotonin	50,397	10,500	0.208	9299.7	0.18
enriched	acetylcholine gaba	15,613,124	3,099,439	0.199	2762765.2	0.17
enriched	acetylcholine serotonin	613,820	93,718	0.153	86533.0	0.12
depleted	octopamine octopamine	1,091	183	0.168	361.6	-0.98
depleted	dopamine serotonin	35,033	3,474	0.099	4197.5	-0.27
depleted	dopamine dopamine	3,193,624	358,505	0.112	424314.7	-0.24
depleted	glutamate glutamate	3,827,209	578,349	0.151	647850.8	-0.16
depleted	dopamine glutamate	622,380	77,558	0.125	86487.2	-0.16

Table 7: Exact neuron type pairs with the largest positive and negative distance-adjusted effects among selected supported pairs.

Direction	Pair	Near pairs	Connected	Fraction	Expected	$\log_2(O/E)$
enriched	T2 cL21	2,776	2,706	0.975	614.7	2.14
enriched	Lawf2 OA-AL2b2	1,160	1,150	0.991	267.1	2.10
enriched	Mi13 cM05	660	659	0.998	169.1	1.96
enriched	Dm10 aMe17c	965	954	0.989	275.3	1.79
enriched	Li31 Tm4	1,480	1,449	0.979	580.1	1.32
depleted	Lawf2 Tm9	6,827	21	0.003	838.9	-5.29
depleted	KCapbp-m KCg-m	71,290	257	0.004	6368.6	-4.63
depleted	Dm3q T1	8,134	20	0.002	505.1	-4.62
depleted	KCab KCapbp-ap1	30,706	84	0.003	2082.0	-4.62
depleted	Dm3q TmY3	7,588	20	0.003	504.6	-4.62

Table 8: Flow pairs with the largest positive and negative distance-adjusted effects among selected supported pairs.

Direction	Pair	Near pairs	Connected	Fraction	Expected	$\log_2(O/E)$
enriched	afferent efferent	322,683	67,261	0.208	39974.4	0.75
enriched	afferent intrinsic	2,271,556	426,365	0.188	263327.5	0.70
enriched	afferent afferent	429,667	62,545	0.146	60821.0	0.04
enriched	intrinsic intrinsic	72,131,799	11,906,258	0.165	12038589.3	-0.02
enriched	efferent efferent	211,888	37,440	0.177	43177.7	-0.21
depleted	efferent intrinsic	2,146,760	312,635	0.146	366243.4	-0.23
depleted	efferent efferent	211,888	37,440	0.177	43177.7	-0.21
depleted	intrinsic intrinsic	72,131,799	11,906,258	0.165	12038589.3	-0.02
depleted	afferent afferent	429,667	62,545	0.146	60821.0	0.04
depleted	afferent intrinsic	2,271,556	426,365	0.188	263327.5	0.70

Table 9: Side pairs with the largest positive and negative distance-adjusted effects among selected supported pairs.

Direction	Pair	Near pairs	Connected	Fraction	Expected	$\log_2(\text{O/E})$
enriched	na right	2,702	494	0.183	368.4	0.42
enriched	left right	7,951,633	1,472,387	0.185	1108888.6	0.41
enriched	left na	2,792	467	0.167	382.6	0.29
enriched	center left	157,811	31,235	0.198	27022.9	0.21
enriched	center right	160,597	31,292	0.195	27428.8	0.19
depleted	left left	34,000,292	5,462,670	0.161	5658493.1	-0.05
depleted	right right	35,235,592	5,813,227	0.165	5988787.9	-0.04
depleted	center center	2,864	731	0.255	753.0	-0.04
depleted	center right	160,597	31,292	0.195	27428.8	0.19
depleted	center left	157,811	31,235	0.198	27022.9	0.21

decline from 44.7% in the nearest bin to 0.89% in the farthest bin, the 27.2 fold contrast between the close and far aggregate windows, and the fitted odds multiplier per 100 nm. The residual and partition analyses show that distance does not explain all structure.

The distance-adjusted enrichments identify metadata partitions whose labels preserve residual connectivity structure after controlling for the global distance distribution. Exact neuron type and brain-region labels show the widest selected effects, which is expected if cell identity and regional circuit architecture impose specific partner preferences beyond physical opportunity. Neurotransmitter labels show smaller selected effects, which is consistent with neurotransmitter being a broad physiological class rather than a precise partner-identity label.

A practical rule for downstream analysis is to treat 0 to 500 nm as a high-opportunity zone and 1.5 to 2.0 microns as a low-opportunity zone under this mesh sampling definition. The high-opportunity zone is connected at 31.3%, while the low-opportunity zone is connected at 1.15%. This does not mean distance alone should be used as a classifier. Instead, it provides a baseline expectation against which region, cell type, neurotransmitter, flow, and side effects can be evaluated.

4 Limitations

The mesh-distance variable is computed from sampled vertices near postsynaptic sites, not from a complete continuous surface-to-surface distance between all dendritic compartments. The 250 nm sampling interval and 500 nm postsynaptic-site radius make the computation tractable at whole-brain scale, but they also define the resolution and biological interpretation of the distance measure.

The report uses the binary question of whether any synaptic connection exists between a near pair. Direction, synapse count, and sign are retained in the enriched table and in the strength summaries, but most partition effects use the binary connected-any outcome. Exact neuron-type plots are selected for support and effect visibility, so the figure is an interpretable high-support view rather than a complete plot of every type pair.

The observed over expected adjustment controls for the global distance-bin distribution but does not control simultaneously for every possible confounder, such as neuron size, synapse count, sampling density, or nested cell-type structure. Claims in this report should therefore be read as descriptive whole-connectome statistics and as hypothesis-generating evidence for circuit specificity.

5 Reproducibility and Outputs

The full enriched near-pair table is stored as `near_pairs_enriched.parquet` in the canonical run directory. The report directory contains the compiled PDF, LaTeX source, figure PDFs, CSV tables, and a JSON manifest with all headline metrics. Lightweight CSV tables are suitable for versioned snapshots. Bulk Parquet tables and mesh caches are intentionally excluded from the repository snapshot.